LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Bernoulli's theorem for compressible and incompressible Flow
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Bernoulli's theorem for compressible and incompressible Flow. Equation of continuity, Lift generation on aerofoil. Reynolds number.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of steady streamlined flow pattern over object in smoke tunnel; Safety: Ensure tunnel is off before entering; Supervision: Instructor monitors flow stability and trainee safety.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the definition of an incompressible fluid?
(b) How does atmospheric pressure change with altitude?
(c) Define the boundary layer in aerodynamics.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Equation of continuity derivation • Bernoulli's theorem for incompressible flow • Bernoulli's theorem for compressible flow • Assumptions in Bernoulli's equation	L & Ds	PPT, IB	15 min
(b)	• Lift generation on aerofoil using Bernoulli • Reynolds number definition and significance • Flow regimes: laminar and turbulent • Boundary layer separation and stall	L & Ds	PPT, PDF	20 min
(c)	• Practical demonstration of smoke tunnel flow • Observation of streamline patterns • Identification of boundary layer effects • Correlation with theoretical predictions	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how Bernoulli's principle applies to lift generation on a helicopter blade.
(b) Differentiate between compressible and incompressible flow assumptions.
(c) Describe the role of the Reynolds number in determining flow regime.

12. Home Assignment and Follow Up	Prepare a brief note comparing the application of Bernoulli's theorem in subsonic versus supersonic flight regimes, referencing the next lesson plan on practical demonstrations.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Practical demonstration of steady streamlined flow pattern over object in a smoke tunnel.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min